

Controllable Visual Synthesis via Diffusion and Flow-Based Generative Models: A Comprehensive Survey

1 Introduction

The landscape of visual synthesis has undergone a paradigmatic shift, transitioning from the stochastic, unconditional generation characteristic of early Generative Adversarial Networks (GANs) [1] to the precise, user-directed control enabled by modern diffusion and flow-based models. While foundational works like [2] and [3] established the tractability of exact likelihood estimation and deterministic sampling in flow-based approaches, the recent ascendancy of diffusion models has redefined the boundaries of generative fidelity. However, the initial promise of text-to-image diffusion models often lacked the fine-grained spatial and structural adherence required for professional applications. This subsection elucidates the evolution toward controllable visual synthesis, where the integration of heterogeneous condition signals—ranging from edge maps and depth sensors to semantic layouts—has become paramount. The introduction of architectures such as [4] marked a critical inflection point, demonstrating that locking pre-trained backbones while injecting spatial controls via zero-initialized convolutions could achieve unprecedented alignment without compromising generative diversity.

Defining the scope of controllability requires a rigorous categorization of input modalities and the mechanisms by which they influence the generative trajectory. Contemporary approaches diverge into architectural modifications, such as the unified frameworks proposed in [5] and [6], which accommodate multiple condition types within a single model, and inference-time guidance strategies that manipulate attention maps or optimize latent trajectories without retraining [7; 8]. A fundamental challenge persists in balancing the triad of fidelity, diversity, and adherence; overly rigid control can collapse the output distribution, while loose guidance fails to capture complex user intents. Theoretical advancements bridging stochastic differential equations (SDEs) and ordinary differential equations (ODEs) through flow matching [9] and rectified flows [10] offer a pathway to resolve these trade-offs by learning straighter integration paths that facilitate more stable and efficient conditional sampling.

Furthermore, the extension of control principles beyond static 2D imagery into video synthesis [11; 12] and 3D asset creation underscores the need for spatiotemporal consistency and multi-view coherence. As models increasingly incorporate discrete state-space formulations for structured data [13; 14], the definition of controllability expands to include logical constraints and physical plausibility. This survey systematically examines these developments, unifying the perspectives of diffusion and flow-based paradigms to provide a comprehensive taxonomy of architectural innovations, guidance mechanisms, and application domains. By critically analyzing the theoretical underpinnings of dynamical systems in generation and evaluating the efficacy of various control injection strategies, we aim to identify persistent gaps in current methodologies and chart a course for future research focused on causal reasoning, neuro-symbolic integration, and the democratization of high-fidelity creative tools.

2 Theoretical Foundations and Unified Dynamical Systems

2.1 Stochastic Differential Equations and the Diffusion Paradigm

The diffusion paradigm fundamentally reconceptualizes image generation as a stochastic dynamical system governed by Stochastic Differential Equations (SDEs), framing the synthesis process as the time-reversal of a gradual data corruption trajectory. This theoretical framework posits a forward

process that continuously perturbs data $\mathbf{x}$ into a known prior distribution, typically Gaussian noise, through a time-dependent SDE defined as $d\mathbf{x} = \mathbf{f}(\mathbf{x}, t)dt + g(t)d\mathbf{w}$, where $\mathbf{f}$ represents the drift coefficient, g the diffusion coefficient, and $d\mathbf{w}$ the standard Wiener process [15]. The generative capability emerges from solving the corresponding reverse-time SDE, which necessitates estimating the score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ of the perturbed data distribution at every timestep. This formulation unifies earlier discrete approaches, such as Denoising Diffusion Probabilistic Models (DDPMs) [16], and continuous score-based methods into a single coherent mathematical language, allowing for flexible noise schedules and sampling strategies.

The core engine driving this reconstruction is the score matching objective, which trains neural networks to approximate the gradient of the log-density of noisy data. By leveraging Denoising Score Matching with annealed noise schedules, models learn to denoise samples across varying levels of corruption, effectively learning the vector field that guides noise back to the data manifold [15; 17]. A critical insight provided by this SDE perspective is the existence of an equivalent Probability Flow Ordinary Differential Equation (ODE) that shares the same marginal distributions as the stochastic process. While the SDE implementation introduces stochasticity that enhances sample diversity and mode coverage, the deterministic ODE counterpart enables exact likelihood estimation and more efficient sampling via high-order solvers [15; 18; 19]. This duality allows practitioners to trade off between the diversity afforded by stochastic samplers and the computational efficiency of deterministic trajectories.

The design of the noise schedule $g(t)$ plays a pivotal role in balancing generation fidelity with controllability. Variance-preserving (VP) and variance-exploding (VE) SDEs offer distinct inductive biases; VP SDEs maintain bounded signal variance, facilitating stable training, while VE SDEs explore the data space more aggressively [15]. Recent advancements suggest that optimizing these schedules jointly with the model parameters can significantly improve likelihood estimates and convergence rates [20]. Furthermore, the stochastic nature of the reverse process is not merely a sampling artifact but a fundamental component for robust conditional guidance. Techniques such as Classifier-Free Guidance exploit the score function’s sensitivity to conditional inputs, interpolating between conditional and unconditional scores to steer the trajectory toward specific semantic concepts without requiring an external classifier [21; 22].

However, the reliance on iterative numerical integration of SDEs presents computational challenges, prompting innovations in non-linear diffusion processes and learned transformations that straighten the probability flow paths [23; 24]. Emerging research also extends this paradigm beyond Gaussian noise, exploring deterministic degradation operators and discrete state spaces to broaden the applicability of diffusion dynamics to inverse problems and discrete data generation [25; 26]. As the field progresses, the theoretical understanding of trajectory regularity and the interplay between stochastic drift and diffusion coefficients remains central to developing faster, more controllable, and physically consistent generative models [27].

2.2 Ordinary Differential Equations and Deterministic Flow Formulations

Complementing the stochastic dynamics detailed previously, the deterministic counterpart to diffusion models is formulated through Ordinary Differential Equations (ODEs) and Continuous Normalizing Flows (CNFs). This paradigm shift from stochasticity to determinism is theoretically grounded in the concept of the probability flow ODE, which defines a unique trajectory that preserves the marginal probability density of the corresponding SDE at every time step [15]. As highlighted in the preceding discussion, this theoretical bridge connects stochastic sampling with deterministic

formulations, enabling exact likelihood computation and deterministic sampling—critical features for applications requiring precise invertibility, such as image editing and latent space interpolation [17]. Unlike the stochastic reverse process which introduces randomness at each integration step, the probability flow ODE allows for the reconstruction of the initial noise vector from a generated sample via simple numerical integration, facilitating bidirectional mapping between the data manifold and the prior distribution [18].

Continuous Normalizing Flows provide the architectural backbone for these deterministic formulations by learning invertible transformations that map a simple base distribution to a complex data distribution. Early innovations like [2] established the viability of flow-based models for high-fidelity image synthesis through tractable log-likelihood optimization. Subsequent advancements, including [3] and [28], addressed limitations in expressivity and density estimation by refining dequantization strategies and incorporating masked convolutions. Furthermore, [29] generalized invertible operations to $d \times d$ convolutions, significantly enhancing spatial modeling capabilities. Despite these successes, traditional CNFs often suffer from computational bottlenecks during training due to the requirement of solving ODEs repeatedly. To mitigate this, [30] introduced multi-scale architectures based on wavelets, enabling efficient modeling of high-resolution images, while [31] augmented the data space to increase model expressivity without prohibitive computational costs.

Building upon these foundational flow architectures, a pivotal development in unifying diffusion and flow perspectives is the emergence of Flow Matching and Rectified Flow frameworks. [9] proposed a simulation-free training objective that regresses vector fields of fixed conditional probability paths, subsuming diffusion paths as specific instances while allowing for more efficient optimal transport (OT) paths. This approach reduces the complexity of the generative trajectory, leading to faster sampling and improved generalization. Building on this, [10] demonstrated that learning ODEs to follow straight paths between source and target distributions minimizes transport cost and enables accurate simulation with coarse time discretization. The theoretical efficacy of these deterministic paths is further supported by [32], which provides rigorous convergence guarantees for fully deterministic sampling under regularity conditions. Additionally, [33] introduced velocity parameterization and variance reduction techniques to enhance the likelihood performance of diffusion ODEs, bridging the gap with state-of-the-art likelihood-based models.

The deterministic nature of these ODE-based formulations offers distinct advantages for controllable generation, particularly in solving inverse problems and performing precise edits. [34] leverages the differentiability of the flow to optimize source noise points, effectively projecting gradients onto the data manifold for controlled synthesis. Similarly, [35] utilizes the simplicity of flow matching to solve linear inverse problems without task-specific tuning. However, the rigidity of deterministic trajectories can sometimes limit diversity compared to stochastic counterparts. Recent works like [36] suggest that reintroducing stochasticity can improve editing fidelity, highlighting a nuanced trade-off between the stability of ODEs and the diversity of SDEs. Future research directions are increasingly focused on hybrid approaches, such as [37], which unifies diffusion and Poisson flow models by varying the dimensionality of the augmented space, offering a continuous spectrum between robust stochastic dynamics and rigid deterministic flows. These developments set the stage for the subsequent exploration of how Flow Matching and Optimal Transport further optimize these trajectories for enhanced controllability and efficiency.

2.3 Unification via Flow Matching and Optimal Transport Theory

The convergence of diffusion probabilistic models and continuous normalizing flows (CNFs) has catalyzed a paradigm shift in generative modeling, unified under the theoretical framework of Flow Matching (FM) and Optimal Transport (OT). While traditional diffusion models rely on stochastic differential equations (SDEs) with fixed Wiener processes, and CNFs traditionally require computationally expensive simulation-based training, FM offers a simulation-free objective that regresses vector fields along arbitrary conditional probability paths [9]. This formulation subsumes diffusion paths as specific instances while enabling the learning of deterministic Ordinary Differential Equations (ODEs) that bridge source and target distributions with unprecedented flexibility. A critical advancement in this unification is the integration of OT theory, specifically the Benamou-Brenier dynamic formulation, to enforce straight-line trajectories between noise and data. By minimizing the kinetic energy of the transport map, OT-based flow matching constructs geodesics that significantly reduce the curvature of integration paths, thereby enhancing sampling efficiency and numerical stability [38; 39].

The theoretical elegance of this approach lies in its ability to transform arbitrary couplings into deterministic flows through iterative rectification. Rectified Flow algorithms demonstrate that recursively learning straight paths between empirical distributions provably decreases convex transport costs, allowing high-fidelity generation with coarse time discretization or even single-step Euler integration [38]. This trajectory straightening is pivotal for controllable image synthesis, as linear paths minimize the accumulation of discretization errors that often distort semantic alignment in complex conditional tasks. Furthermore, the unification extends to latent spaces, where Flow Matching in pretrained autoencoder manifolds reduces computational overhead while preserving the geometric properties necessary for precise editing and interpolation [40; 41]. The versatility of this framework is further evidenced by its adaptation to discrete state-spaces via Continuous Time Markov Chains, effectively bridging the gap between continuous flow dynamics and discrete diffusion processes for applications in language and molecular design [38; 42].

Despite these advancements, challenges remain in balancing the rigidity of straight paths with the diversity guaranteed by stochastic perturbations. While deterministic OT paths offer speed, they may occasionally suffer from mode collapse if the underlying coupling is not sufficiently expressive. Recent works address this by introducing stochastic variants or hybridizing FM with classifier-free guidance strategies to recover diversity without sacrificing the efficiency of straight trajectories [43; 44]. Moreover, the application of piecewise rectification allows for local straightening of trajectories, offering a middle ground that accommodates complex data manifolds while maintaining fast inference [45]. Looking forward, the synthesis of FM with stochastic optimal control promises to refine these dynamics further, enabling memoryless noise schedules that optimize reward functions directly along the flow [46]. Ultimately, the unification via FM and OT establishes a robust mathematical foundation where the trade-off between sampling speed, likelihood estimation, and controllability can be systematically navigated, paving the way for next-generation models that are both theoretically grounded and practically efficient for high-dimensional visual synthesis.

2.4 Advanced Dynamical Systems for Inverse Problems and Control

Building upon the robust mathematical foundation of Flow Matching and Optimal Transport established previously, this section transitions from optimizing intrinsic trajectory efficiency to exercising active dynamical control over the generative process. While the preceding discussion highlighted how straight-line paths enhance sampling speed and stability, addressing complex inverse problems

and enforcing rigid constraints necessitates a shift from passive generation to steering these dynamics via physical laws, sparse measurements, and reward signals. A primary paradigm in this domain frames guidance as a stochastic optimal control problem, where the generative trajectory is modified by optimizing a specific cost functional. Recent work by [47] exemplifies this by utilizing adjoint methods to derive drift adjustments that align outputs with reference styles through terminal costs, effectively decoupling content and style without retraining. Similarly, [48] proposes differentiating through the entire denoising chain to optimize low-rank adapters against differentiable reward functions, offering a principled alternative to high-variance reinforcement learning.

Beyond reward optimization, solving inverse problems often requires approximating posterior distributions conditioned on partial observations. Traditional approaches that simply replace known pixels during sampling can lead to incoherence; instead, rigorous Bayesian frameworks modify the reverse drift to sample from the true posterior. [49] integrates differentiable forward operators directly into the denoising step, enabling the generation of 3D scenes or other latent signals consistent with 2D projections without direct supervision. This concept is further refined in [50], which recasts sampling as a constrained optimization problem, employing iterative projections to ensure generated trajectories adhere to non-convex physical constraints or morphometric properties. Complementing this, [51] introduces relative trajectory balance, a data-free objective derived from generative flow networks that enables unbiased sampling from arbitrary posteriors defined by black-box constraints.

The flexibility of these dynamical systems is further enhanced by generalizing the underlying corruption process itself. Moving beyond standard Gaussian noise, [25] establishes that generative behavior persists even with deterministic degradations like blurring or masking, allowing the inversion of arbitrary image transforms. This aligns with [52], which learns time-dependent parameters for nonlinear reaction-diffusion equations to solve specific restoration tasks efficiently. Furthermore, leveraging the deterministic nature of flow-based perspectives offers distinct advantages for control; [34] exploits the invertibility of probability flow ODEs to optimize the source noise vector directly, projecting gradients onto the data manifold for precise control in linear and non-linear inverse tasks. Theoretical guarantees for such deterministic approaches are strengthened by [44], which addresses the off-manifold phenomena of standard classifier-free guidance by reformulating it as a manifold-constrained inverse problem, thereby improving invertibility and reducing mode collapse. Collectively, these advanced dynamical systems signify a maturation of the field from purely probabilistic modeling to hybrid physics-informed and control-theoretic frameworks, paving the way for generative models that are not only creative but also rigorously compliant with external constraints and observable data.

3 Architectural Paradigms for Multimodal Condition Integration

3.1 Adapter-Based Architectures for Spatial and Structural Control

Adapter-based architectures represent a pivotal paradigm shift in controllable image synthesis, enabling the injection of complex spatial and structural conditions into frozen, pre-trained diffusion backbones without compromising their intrinsic semantic knowledge or requiring full-model retraining. This approach addresses the critical challenge of aligning high-fidelity generative priors with precise user intents, such as edge maps, depth sensors, and pose skeletons, through lightweight, plug-in modules. The foundational innovation in this domain is exemplified by ControlNet, which introduces a “zero convolution” mechanism—layers initialized with zero weights that progressively learn to modulate feature maps while initially nullifying their impact to prevent noise injection into

the robust backbone [4].

Building upon this zero-initialization strategy, recent advancements have focused on enhancing training stability and convergence speed. ControlNeXt proposes replacing zero convolutions with Cross Normalization (CN), a mechanism that achieves rapid and stable convergence while reducing learnable parameters by up to 90% compared to traditional adapter branches [53]. Furthermore, to address the limitation of strict pixel-perfect alignment which can constrain creativity, ControlNet++ introduces an efficient consistency feedback loop. By optimizing pixel-level cycle consistency between generated outputs and input conditions using a single-step denoising reward strategy, it significantly improves adherence to structural constraints like segmentation masks and depth maps without the prohibitive computational cost of multi-step gradient backpropagation [54].

The scope of adapter-based control has also expanded beyond single-modality constraints to support unified, composable conditioning. Uni-ControlNet demonstrates that a single framework can simultaneously handle diverse local controls (e.g., edges, depth) and global controls (e.g., image embeddings) by fine-tuning only two additional adapters, thereby facilitating flexible compositionality regardless of the number of input conditions [5]. Similarly, Composer decomposes images into representative factors, training diffusion models to recompose inputs from composable elements like sketches and color histograms, effectively creating a vast design space for customizable content creation [55]. For scenarios requiring resolution flexibility, ResAdapter serves as a domain-consistent module that allows frozen models to generate images at unrestricted resolutions and aspect ratios by learning pure resolution priors, proving compatible with other control adapters [56].

In the temporal domain, these architectural principles extend to video synthesis, where maintaining structural consistency across frames is paramount. MOFA-Video employs domain-aware motion field adapters to generate dense motion flows from sparse controls like human landmarks, wrapping multi-scale features to guide frozen image-to-video models for stable animation [57]. Additionally, extending spatial control to three dimensions, SyncDreamer utilizes 3D-aware feature attention mechanisms to synchronize intermediate states across multiple views, ensuring geometric consistency in novel view synthesis tasks [58]. Despite these successes, challenges remain in balancing the trade-off between strict structural adherence and generative diversity, particularly when conditions are sparse or ambiguous. Future directions likely involve integrating neuro-symbolic reasoning into adapter modules to better interpret semantic intent behind structural constraints, moving from rigid geometric locking to context-aware structural guidance.

3.2 Attention-Centric Modulation for Semantic and Layout Alignment

Attention-centric modulation has emerged as a pivotal architectural paradigm for achieving fine-grained semantic and spatial controllability, directly addressing the limitations of adapter-based methods that operate orthogonally to a model’s internal reasoning. Unlike the auxiliary network injections discussed previously, this strategy manipulates the intrinsic cross-attention and self-attention mechanisms of transformer-based diffusion models to enforce precise alignment between textual prompts and visual layouts. The fundamental premise relies on the observation that cross-attention maps inherently encode the spatial localization of semantic concepts, allowing for direct intervention without extensive retraining. Early works demonstrated that rewriting or guiding these attention scores using spatial masks or bounding boxes could effectively bind text tokens to specific image regions, thereby mitigating attribute leakage and enabling layout-to-image synthesis [59; 7]. By treating attention maps as editable interfaces, methods such as [60] introduce optimization objectives to activate specific regions in cross-attention maps, granting users high-level

positional control over multiple objects while preserving background coherence.

To address the challenge of complex scene composition involving multiple entities, recent advancements have focused on decomposing attention mechanisms across spatial and temporal dimensions. [61] proposes explicit control over both spatial regions and denoising timesteps, optimizing combination weights to ensure that objects appear in correct locations with appropriate attributes. Furthermore, the issue of semantic interference in multi-subject generation has been tackled through bounded attention strategies that restrict information flow between distinct subject tokens, preventing the blending of visual features [62; 63]. These methods often employ localized supervision or masking during the sampling process to enforce strict separation of semantic manifolds. Beyond cross-attention, self-attention modulation plays a critical role in maintaining structural consistency and style transfer. Techniques like mutual self-attention allow the model to query correlated textures from reference images, facilitating tuning-free consistent generation and non-rigid editing [8]. Similarly, swapping keys and values in self-attention layers enables effective visual style prompting while adhering to content constraints [64].

Emerging trends integrate energy-based models and Bayesian inference into attention layers to dynamically update context vectors, enhancing the handling of ambiguous or relational descriptions [65; 66]. For instance, [66] formulates an object-centric attribute binding loss derived from an energy-based model to correct misalignments without fine-tuning. Additionally, the integration of Large Language Models (LLMs) to generate coarse layouts or programmatic sketches further refines attention guidance, bridging the gap between high-level reasoning and pixel-level synthesis [67; 68]. Despite these successes, challenges remain in balancing the rigidity of attention constraints with the stochastic nature of diffusion, as overly aggressive modulation can degrade image fidelity or introduce artifacts. Future directions likely involve hierarchical attention decomposition that separates global scene structure from local details, alongside adaptive mechanisms that learn to modulate attention intensity based on prompt complexity. While these attention-centric techniques offer precise manipulation of internal representations, they often function as modifications to existing architectures rather than holistic solutions. This limitation motivates the exploration of unified multi-modal transformer architectures that shift from modular interventions to end-to-end frameworks natively ingesting heterogeneous conditions, a paradigm that will be explored in the following section.

3.3 Unified Multi-Modal Transformer Architectures

Moving beyond the modularity of adapter-based systems, unified multi-modal transformer architectures represent a paradigm shift toward end-to-end frameworks that natively ingest heterogeneous conditions—spanning text, sketches, reference images, and audio—within a single joint distribution. This holistic approach eliminates the semantic friction often inherent in stitching together disparate encoders and frozen backbones, instead fostering bidirectional information flow across modalities through a shared latent space. A foundational strategy in this domain involves tokenizing diverse inputs into a unified sequence, enabling the transformer to learn cross-modal interactions directly. For instance, [69] projects reference images into pseudo-word embeddings within a unified latent space, allowing joint text-image conditioning to guide generation without the need for separate injection mechanisms. Similarly, [70] standardizes heterogeneous generation intents via multi-modal instructions, amalgamating disparate modalities like edges and styles into a natural language format that fine-tunes a diffusion model to generalize across unseen tasks.

The scalability of these unified architectures is significantly enhanced by integrating flow match-

ing objectives with large-scale Diffusion Transformers (DiTs). [71] demonstrates that employing separate weights for image and text modalities while enabling bidirectional flow improves text comprehension and typography, adhering to predictable scaling laws. This trajectory culminates in frameworks like [72], which utilizes learnable placeholder tokens (e.g., [53]) and Rotary Positional Embeddings (RoPE) to unify representations across spatial-temporal resolutions, supporting the generation of images, videos, and 3D objects within a single flagship model. Further advancing this flexibility, [73] conceptualizes images as sequences of dynamically-sized tokens, resolving resolution biases and enabling unrestricted aspect ratio generation, a critical capability for universal synthesis engines.

Beyond static image synthesis, unified architectures are redefining spatiotemporal control. [74] introduces a multimodal video block that decouples cross-attention layers to simultaneously process image and text conditions for appearance, while optionally integrating geometric controls via pre-trained modules. In the realm of composition, [55] decomposes scenes into representative factors (e.g., depth, palette, layout) treated as composable elements within a unified diffusion process, exponentially expanding the design space for customizable creation. Furthermore, [75] pushes the boundary of modality agnosticism by aligning input and output spaces to generate any combination of modalities from any input set, even those absent in training data, effectively bridging the gap between unimodal specialists and generalist agents.

Despite their promise, these unified models face challenges in computational complexity and training stability when scaling to high-dimensional multi-modal spaces. Emerging solutions leverage state-space models to mitigate quadratic attention costs; [76] and [77] illustrate how replacing attention with linear-complexity State Space Models preserves expressive power while enabling efficient high-resolution synthesis. Looking forward, the convergence of unified tokenization, flow-based dynamics, and efficient sequence modeling suggests a trajectory toward universal generative interfaces capable of seamless, real-time interaction across all sensory modalities, fundamentally altering the landscape of controllable visual synthesis.

3.4 Disentangled Representations and Dynamic Feature Modulation

Building upon the unified multi-modal frameworks discussed previously, which excel at ingesting heterogeneous conditions but often result in entangled representations, a critical complementary paradigm focuses on architectural decoupling to achieve fine-grained controllability. While unified models foster bidirectional information flow, explicit architectural interventions are frequently required to independently manipulate semantic attributes such as style, content, and structure without compromising structural integrity. A foundational approach in this domain leverages invertible flows to establish bijective mappings between data and latent spaces, providing a rigorous mathematical framework for precise attribute editing. [78] formulates conditional exploration as a continuous normalizing flow within the GAN latent space, facilitating attribute-conditioned sampling that disentangles factors like pose and illumination while preserving identity. Similarly, [79] introduces conditional INNs that combine generative flows with feed-forward conditioning networks, ensuring diverse sampling and sharp image reconstruction without the mode collapse typical of adversarial training, though these methods often necessitate specific architectural constraints to maintain invertibility.

Complementing flow-based decoupling, dynamic feature modulation via adaptive normalization and hypernetworks has emerged as a potent mechanism for injecting conditions without altering backbone weights, effectively bridging the gap between frozen backbones and flexible control. [6]

employs a task-aware HyperNet to dynamically generate modulation parameters for a frozen diffusion backbone, enabling a single model to handle diverse visual conditions ranging from depth maps to sketches. This dynamic weight generation allows the network to adapt its feature extraction pathways based on the input modality, effectively separating structural guidance from the semantic generation process. Furthermore, [80] addresses the trade-off between style transfer and text controllability by utilizing Q-Formers to decouple style and semantic representations, injecting them into mutually exclusive subsets of cross-attention layers. This architectural separation ensures that style injection does not degrade the model’s adherence to textual prompts, a common failure mode in less structured approaches.

Recent advancements have further refined these strategies by disentangling global and local representations through sophisticated attention mechanisms and layered scene decompositions. [81] proposes a feature-space decoupling mechanism where style and content are separated via simple addition and subtraction operations, injecting reference features exclusively into style-specific blocks to prevent content leakage. This contrasts with [82], which optimizes a layered scene representation during sampling to achieve spatial disentanglement, allowing for independent movement and resizing of objects in generated scenes. Additionally, [83] reveals that the posterior mean predictor in diffusion models operates within low-dimensional semantic subspaces, enabling training-free editing directions that are homogeneous and transferable.

Despite these advances, challenges remain in balancing the degree of disentanglement with generation fidelity; over-constraining latent factors can lead to rigid outputs lacking diversity, while insufficient separation results in attribute bleeding. Future research directions point towards neuro-symbolic integration, where explicit logical rules guide the disentanglement process, and the unification of flow matching objectives with dynamic modulation to create straighter, more interpretable latent trajectories. The synthesis of invertible flows, dynamic hypernetworks, and attention-based decoupling represents a critical evolution in architectural paradigms, moving from monolithic generation towards modular, composable, and precisely controllable visual synthesis, setting the stage for the next frontier in generative modeling.

4 Guidance Mechanisms and Inference-Time Control Strategies

4.1 Gradient-Based Guidance and Optimal Control Formulations

Gradient-based guidance and optimal control formulations represent a pivotal paradigm in steering the generative trajectories of diffusion and flow-based models without necessitating weight updates. These methods fundamentally operate by manipulating the score function or the underlying vector field during the sampling phase, effectively biasing the reverse process toward regions of the data manifold that satisfy external constraints, semantic attributes, or reward objectives. The foundational mechanism, classifier guidance, integrates gradients from an auxiliary classifier into the reverse-time stochastic differential equation (SDE) to enhance class conditional fidelity [15]. However, the reliance on separately trained classifiers introduces potential gradient misalignment; recent advancements address this by employing adversarially robust classifiers, which yield gradients more consistent with human perception and significantly improve generation quality [22]. To circumvent the need for external classifiers, classifier-free guidance interpolates between conditional and unconditional score estimates, offering a robust trade-off between sample diversity and adherence to prompts [21]. Theoretical analyses indicate that while strong guidance tightens alignment with task-specific information, it inherently reduces the differential entropy of the output distribution, potentially leading to mode collapse [84].

Beyond standard classification, energy-based models (EBMs) provide a flexible framework for enforcing complex geometric, physical, or semantic rules by defining generative tasks as energy minimization problems. By differentiating external loss functions, these approaches project the sampling trajectory onto feasible manifolds, enabling compositional generation and constraint satisfaction that traditional guidance methods struggle to achieve [85; 50]. This perspective unifies various control strategies, allowing diffusion models to serve as plug-and-play priors for solving inverse problems where auxiliary differentiable constraints guide the reconstruction of high-dimensional data [86]. Furthermore, universal guidance algorithms have emerged to control pre-trained models with arbitrary modalities, such as segmentation or object detection signals, by leveraging gradient-based updates without retraining task-specific components [87].

The formulation of guidance as a stochastic optimal control (SOC) problem offers a rigorous mathematical framework for reward fine-tuning and trajectory optimization. Approaches like Adjoint Matching cast reward maximization as a control problem governed by Hamilton-Jacobi-Bellman equations, utilizing memoryless noise schedules and adjoint methods to optimize generation paths while preserving sample diversity [46]. Similarly, Reference-Based Modulation employs SOC to decouple style and content from reference images, encoding desired attributes through terminal costs to ensure high-fidelity personalization without additional adapters [47]. In the context of flow-based models, differentiating through the flow allows for the direct optimization of source noise points, effectively projecting gradients onto the data manifold to solve linear and non-linear controlled generation tasks [34]. Despite their efficacy, these gradient-based methods face challenges regarding computational overhead and the stability of backpropagation through long sampling chains. Future directions lie in developing more efficient solvers that approximate these optimal control trajectories with fewer function evaluations and in bridging the gap between heuristic guidance scales and theoretically grounded control policies.

4.2 Latent Space Inversion and Trajectory Optimization

While gradient-based guidance and optimal control formulations steer generative trajectories through external constraints, latent space inversion and trajectory optimization offer a complementary paradigm by directly manipulating the internal initialization and path of the reverse process. These inference-time control methods enable the precise editing of pre-trained diffusion and flow models without weight updates, fundamentally relying on the accurate mapping of a real image $\mathbf{x}_0$ back to its corresponding latent noise vector $\mathbf{x}_T$ or intermediate trajectory $\{\mathbf{x}_t\}_{t=0}^T$. This mapping facilitates operations that preserve structural integrity while altering semantic attributes. Early approaches leveraged the deterministic nature of probability flow ordinary differential equations (ODEs) to perform inversion via reverse-time integration; however, standard discretization often accumulates errors, leading to poor reconstruction fidelity. To address this, [17] introduced non-Markovian processes that share the same training objective as stochastic counterparts but allow for significantly faster, deterministic sampling and inversion, establishing a critical trade-off between stochasticity and reconstruction accuracy.

Recent advancements have shifted towards optimization-based inversion strategies to overcome the limitations of direct numerical solvers. A seminal contribution in this domain is [88], which proposes optimizing the unconditional text embedding rather than the noise vector itself. This “null-text” optimization effectively decouples the inversion process from the specific prompt, allowing for high-fidelity reconstruction and robust text-guided editing. Complementing this, [89] argues that the native Gaussian noise space lacks a convenient structure for editing. By proposing an alternative latent space where noise maps are not statistically independent across timesteps, this

work enables meaningful manipulations such as shifting and color edits while maintaining perfect reconstruction. Further refining the inversion landscape, [90] demonstrates that disentangling the source and target diffusion branches during inversion yields superior content preservation and edit fidelity compared to unified optimization schemes, achieving state-of-the-art results with minimal computational overhead. Similarly, [91] addresses the loss of spatial context in existing methods by introducing a guidance mechanism that preserves local structural details without requiring per-timestep optimization.

Beyond static image inversion, trajectory optimization has emerged as a vital mechanism for refining generative paths to satisfy complex constraints, bridging the gap between the optimal control theories discussed previously and practical editing applications. In the context of flow-based models, [9] and [10] demonstrate that learning straighter probability paths via optimal transport not only accelerates sampling but also simplifies the trajectory, making it more amenable to linear interpolation and control. For dynamic tasks, [92] explores compositional diffusion priors, utilizing inference-time mechanisms like “DoubleTake” to stitch sequential motion clips, effectively optimizing the trajectory to ensure smooth transitions between prompted intervals. Furthermore, [46] frames trajectory refinement as a stochastic optimal control problem, proving that enforcing memoryless noise schedules during reward fine-tuning allows for the precise steering of generative paths toward arbitrary reward models while maintaining diversity.

Despite these advances, a persistent tension remains between reconstruction fidelity and editability. Highly accurate inversions often lock the latent code into a rigid manifold that resists semantic modification, whereas looser approximations facilitate editing at the cost of identity preservation. Future research directions point towards adaptive trajectory rectification, where the curvature of the generative path is dynamically adjusted based on the specific editing task, and the unification of discrete and continuous flow matching [38; 93] to extend these powerful inversion capabilities to non-continuous domains such as language and molecular graphs. While these methods refine the generative path from a global perspective, the following section explores a complementary, fine-grained approach: manipulating attention mechanisms within transformer layers to achieve precise spatial and semantic alignment without necessitating parameter updates.

4.3 Attention Mechanism Manipulation and Feature Injection

Attention mechanism manipulation represents a pivotal paradigm in inference-time control, enabling precise spatial and semantic alignment by directly intervening in the transformer layers of diffusion and flow models without necessitating parameter updates. Unlike adapter-based architectures that require auxiliary networks, these training-free strategies leverage the intrinsic interpretability of cross-attention and self-attention maps to enforce user constraints dynamically. The core premise posits that cross-attention scores fundamentally dictate the spatial correspondence between textual tokens and visual regions, while self-attention layers govern structural coherence and texture preservation. By modulating these attention distributions during the denoising trajectory, one can achieve fine-grained layout control, robust concept binding, and consistent identity maintenance.

A primary application of this methodology is spatial layout enforcement, where user-specified bounding boxes or segmentation masks guide the generation process. Techniques such as [7] and [94] introduce spatial constraints directly into the attention computation, penalizing deviations between attention maps and target regions. Specifically, [94] formulates inner-box, outer-box, and corner constraints to ensure objects appear strictly within designated areas, effectively mitigating

the “attribute leakage” common in unconditional generation. Similarly, [95] proposes a region and boundary-aware guidance approach that refines attention maps discretely, enhancing object discriminability and layout fidelity. Advanced methods like [96] utilize large language models to automatically infer layouts, subsequently applying refocusing losses to align attention peaks with semantic entities. Furthermore, [97] employs a predicted BoxNet to generate unique masks for each entity, constraining cross-attention to prevent semantic entanglement in complex scenes.

Beyond spatial arrangement, attention manipulation is critical for maintaining consistency in multi-subject generation and video synthesis. In multi-subject scenarios, standard diffusion models often suffer from feature blending, where distinct identities merge. [98] introduces bounded attention to restrict information flow between subjects, preserving individual characteristics even in crowded compositions. Extending this to personalized generation, [45] utilizes localized attention supervision to ensure reference subjects attend only to their corresponding regions, while [99] orchestrates multi-subject cross-attention to harmonize distinct identities within a single frame. For temporal consistency, [100] propagates diffusion features across video frames by enforcing consistency in the attention space, effectively reducing flicker without retraining. [11] further enhances this by integrating fully cross-frame interaction within self-attention modules, ensuring appearance coherence throughout long sequences. In the domain of style transfer and editing, [101] converts standard self-attention into mutual self-attention, allowing the model to query textures from a source image to maintain identity during non-rigid edits. Likewise, [64] achieves style consistency by swapping keys and values in late self-attention layers, decoupling style from content without fine-tuning.

Despite their efficacy, these approaches face trade-offs between guidance strength and image naturalness. Aggressive attention masking can lead to artifacts or reduced diversity, as noted in analyses by [102], which highlights that self-attention is crucial for geometric integrity while cross-attention drives semantic placement. Future directions may involve adaptive modulation schemes that dynamically balance constraint adherence with generative freedom, potentially integrating energy-based objectives like those in [66] to optimize attention distributions probabilistically. Ultimately, attention manipulation stands as a versatile, computationally efficient mechanism for bridging the gap between abstract user intent and concrete visual synthesis.

4.4 Plug-and-Play Experts and Modular Guidance Frameworks

Building upon the internal state interventions discussed previously, the paradigm of plug-and-play experts and modular guidance frameworks represents a significant evolution in inference-time control, shifting from monolithic conditioning mechanisms to dynamic ensembles of specialized models. While attention manipulation offers fine-grained control through transformer layers, this approach addresses the inherent limitations of single-guidance strategies by treating pre-trained diffusion or flow models as interchangeable energy-based priors. This enables the composition of diverse expert signals—ranging from semantic layouts to physical constraints—into a unified generative trajectory without catastrophic forgetting or mode collapse. A foundational insight in this domain is the interpretation of diffusion models as score-based energy functions, allowing for the linear superposition of gradients from multiple sources. For instance, [103] demonstrates that distinct diffusion models, each trained on specific object categories or attributes, can be composed at test time to generate scenes with novel combinations of concepts that were never jointly observed during training. This composability extends to heterogeneous modalities; [87] proposes a universal algorithm that integrates arbitrary guidance functions, such as face recognition or object detection scores, without requiring retraining of the base generative model, thereby enabling true plug-and-play adaptability.

To manage the computational overhead and potential conflicts arising from multiple guidance signals, recent advancements have introduced noise-level specialization and parameter-efficient adapters. The concept of assigning specific experts to distinct timesteps leverages the semantic hierarchy inherent in the diffusion process, where early steps determine global structure and later steps refine texture. [104] exemplifies this by deploying specialized networks for different modalities and predicting spatial-temporal influence functions, ensuring robust guidance throughout the reverse process. Furthermore, the integration of lightweight adapters has facilitated the simultaneous utilization of multiple local and global controls. [54] introduces a unified framework where diverse conditions like edge maps, depth, and segmentation are processed through a constant number of adapters, drastically reducing fine-tuning costs while maintaining composability. Similarly, [53] advances this efficiency by replacing heavy auxiliary branches with minimal cross-normalization layers, achieving powerful control with significantly fewer learnable parameters. These modular architectures not only enhance flexibility but also improve stability; [54] optimizes pixel-level cycle consistency using efficient consistency feedback, ensuring that the composed guidance signals remain aligned with the generated output.

Despite these successes, challenges remain in balancing the influence of competing experts and maintaining sampling efficiency. The naive summation of score functions can lead to over-constrained trajectories that sacrifice diversity for fidelity, a trade-off theoretically analyzed in [84]. To mitigate this, stochastic optimal control formulations, such as those in [47], provide a rigorous framework for balancing style and content descriptors via terminal costs, enabling precise decoupling without external adapters. Additionally, the emergence of neuro-symbolic approaches like [105] suggests a future where large language models orchestrate sequences of visual experts, dynamically selecting and arranging plug-and-play modules based on high-level instructions. As the field progresses, the integration of discrete state-space flows [106] and accelerated sampling techniques [45] will be crucial for scaling these modular frameworks to real-time applications, ultimately realizing a vision of generative AI where control is as fluid and composable as the creative intent itself.

5 Advanced Applications in Editing, Personalization, and Dimensional Extension

5.1 Subject-Driven Personalization and Concept Disentanglement

Subject-driven personalization represents a paradigm shift in controllable generation, enabling the binding of unique visual concepts—such as specific identities, artifacts, or artistic styles—to textual identifiers within pre-trained diffusion models. The core challenge lies in achieving high-fidelity preservation of the subject while maintaining the model’s inherent editability and compositionality. Early approaches primarily relied on optimization-based fine-tuning, where [107] introduced a class-specific prior preservation loss to bind a unique token to a subject, effectively recontextualizing it without catastrophic forgetting. While effective, such full-model or extensive parameter updates are computationally prohibitive and prone to overfitting. To address this, parameter-efficient strategies have emerged, focusing on low-rank adaptations. [108] restricts optimization to the singular values of weight matrices, drastically reducing the parameter space and mitigating language drift, while [109] identifies specific low-rank directions corresponding to interpretable attributes, allowing for continuous, disentangled control over concepts like age or expression without interfering with global semantics. Furthermore, [110] theoretically guarantees the preservation of hyperspherical energy, ensuring that the semantic generation capability of the base model remains intact during adaptation.

Parallel to fine-tuning, a significant trend involves training-free inference strategies that leverage pre-

trained encoders to inject subject features dynamically. [111] and [112] utilize multimodal encoders to extract robust subject representations, enabling zero-shot personalization by conditioning the diffusion process on these embeddings rather than updating model weights. This approach offers substantial speedups, as evidenced by [63], which employs localized attention mechanisms and delayed conditioning to prevent identity blending in multi-subject scenarios, achieving generation speeds orders of magnitude faster than optimization-based counterparts. Similarly, [113] constructs a unified framework capable of handling open-domain personalization from single reference images by integrating coarse location and fine-grained reference controls. However, these methods often struggle with the entanglement of identity and background context. To disentangle these factors, [80] employs Q-Formers to decouple style and semantics, injecting them into mutually exclusive cross-attention layers, while [47] frames personalization as a stochastic optimal control problem, using a style descriptor to guide the drift term and separate content from style without additional adapters.

The frontier of this domain addresses the complexity of multi-concept composition and precise attribute control. [114] and [115] highlight the necessity of aligning textual embeddings with visual regions to prevent concept confusion, with the latter proposing contrastive fine-tuning of the text encoder itself to resolve semantic collisions. Moreover, [116] introduces a prompt spectrum space, representing images as collections of inverted token embeddings across different denoising stages to achieve fine-grained attribute disentanglement. Despite these advances, challenges persist in balancing subject fidelity with text alignment, particularly when subjects exhibit complex geometries or when multiple customized concepts must coexist harmoniously [117]. Future research must therefore focus on developing unified latent spaces that inherently disentangle identity, pose, and lighting, potentially leveraging neuro-symbolic approaches to enforce logical constraints on concept composition, thereby moving from mere appearance mimicry to true conceptual understanding and manipulation.

5.2 Precision Image Editing and Layout Manipulation

Bridging the gap between the subject-centric adaptation discussed previously and the dynamic temporal synthesis explored subsequently, precision image editing and layout manipulation represent the pinnacle of user-directed control in generative modeling. This domain transitions from holistic synthesis or static personalization to granular spatial and semantic modification, addressing the fundamental challenge of altering specific image attributes while rigorously preserving the structural integrity and photorealistic consistency of the unedited background. Early breakthroughs established mask-guided semantic inpainting as a foundational paradigm, where methods like [118] and [119] pioneered the seamless fusion of text-guided diffusion latents with noised input versions across progressive noise levels. These approaches effectively constrain edits to user-defined regions of interest (ROIs), ensuring coherent context completion. Building upon this, [120] introduced dual-modality conditioning, integrating shape priors with textual descriptions to achieve precise object insertion without background degradation, while [121] advanced the field by automating mask generation through contrasting conditional predictions, thereby reducing reliance on manual segmentation.

Beyond static masking, recent innovations have shifted towards interactive, point-based manipulation and continuous layout optimization, enabling non-rigid deformations and object repositioning that were previously intractable. The emergence of “drag-style” editing, exemplified by [122] and [123], leverages the rich semantic correspondence within intermediate U-Net features to optimize latent representations. These methods translate user-specified handle points into gradient-based

supervision signals, allowing for intuitive pixel-level control over object posture and geometry. Complementing this, [124] offers a more efficient alternative by editing predicted noise outputs directly in a single denoising step, significantly mitigating optimization latency while preserving original content fidelity. For global structural reconfiguration, [82] proposes a training-free layered scene representation that jointly denoises renderings at varying spatial layouts, facilitating complex operations such as object cloning, resizing, and 3D rotation without model retraining. Similarly, [125] unifies 2D and 3D editing by formulating transformations as geometric constraints embedded within attention layers, ensuring physically plausible lighting and shadow consistency during object manipulation.

Layout manipulation extends these capabilities to multi-object scene composition, where maintaining spatial relationships is paramount. [126] introduces object-aware cross-attention mechanisms to enforce strict adherence to bounding box constraints, effectively resolving the multimodal fusion challenges inherent in complex scenes. Furthermore, training-free strategies like [7] manipulate cross-attention maps directly to steer generation towards user-specified layouts, demonstrating that robust spatial control can be achieved without extensive fine-tuning. Despite these advances, trade-offs remain between edit fidelity and background preservation; aggressive latent optimization can introduce artifacts, as noted in studies comparing inversion techniques [88; 90]. Future directions point towards neuro-symbolic frameworks that integrate explicit geometric reasoning with diffusion priors, as seen in [105], to achieve even higher levels of controllability and logical consistency. Such advancements in precise spatial control provide the essential groundwork for extending these principles into the temporal domain, where maintaining consistency across frames becomes the next critical frontier.

5.3 Spatiotemporal Consistency in Video Synthesis and Animation

Extending controllable generative principles to the temporal domain necessitates a rigorous treatment of spatiotemporal consistency, addressing the dual challenges of maintaining high-fidelity motion coherence and preserving subject identity across frames. Unlike static image synthesis, video generation requires models to capture complex physical dynamics and long-range temporal dependencies while adhering to user-specified constraints. Early approaches often decoupled motion and appearance, utilizing hierarchical frameworks where optical flow guides texture synthesis to ensure plausible dynamics [127]. However, modern diffusion-based paradigms have shifted towards joint modeling of space and time. For instance, [128] extended standard 2D architectures to 3D U-Nets, enabling joint training on image and video data to reduce gradient variance, while [129] introduced a Space-Time U-Net that generates the entire temporal duration in a single pass, effectively mitigating the inconsistency issues inherent in keyframe-based super-resolution pipelines.

A critical frontier in this domain is the injection of explicit control signals to guide motion trajectories and structural evolution. Methods like [12] integrate motion and content priors via adaptive noise initialization strategies based on optical flow, allowing for fine-grained control over object movement. Similarly, [130] factorizes image-to-video synthesis into motion field prediction and feature propagation, enabling precise trajectory control through sparse annotations. For character animation, [131] employs a ReferenceNet to merge detailed appearance features via spatial attention, ensuring identity preservation under drastic pose variations driven by skeletal guides. Complementary to these architectural innovations, training-free mechanisms have emerged as efficient alternatives. [132] reprograms frame-level self-attention into cross-frame attention to enforce background consistency without additional training, while [133] propagates diffusion features based on inter-frame correspondences to maintain coherence during text-driven editing.

Despite these advancements, achieving long-term consistency remains non-trivial due to error accumulation in autoregressive sampling. To address this, [134] proposes a temporal co-denoising paradigm that extends short video models to handle hundreds of frames with diverse semantic segments. Furthermore, [135] leverages state-space models for efficient temporal modeling, facilitating the generation of minute-long videos with stable identity retention. The integration of 3D geometric priors also offers a promising direction for enhancing consistency. [136] utilizes monocular depth and video-diffusion priors to jointly synthesize multi-view consistent frames, reducing drift in camera-controlled generation. Likewise, [137] employs epipolar attention mechanisms to synchronize multiple videos rendered from different camera trajectories, ensuring geometric fidelity across views.

Looking forward, the convergence of explicit motion modeling with implicit generative priors represents a pivotal trend. While methods like [138] demonstrate that parameter-free temporal shifts can effectively capture motion dynamics, future research must prioritize the unification of controllability and efficiency. The development of robust metrics for quantifying temporal coherence and the exploration of neuro-symbolic approaches to enforce physical laws within the diffusion process will be essential for advancing toward truly controllable, high-fidelity video synthesis systems.

5.4 Three-Dimensional and Four-Dimensional Content Generation

Building upon the integration of 3D geometric priors for video consistency discussed previously, the extension of diffusion and flow-based generative models to full three-dimensional (3D) and four-dimensional (4D) content creation represents a paradigm shift in geometric modeling. This domain moves from explicit mesh construction to implicit neural field optimization guided by 2D priors. A foundational approach in this transition is Score Distillation Sampling (SDS), which leverages pretrained 2D text-to-image diffusion models as density estimators to optimize parametric 3D representations, such as Neural Radiance Fields (NeRFs), without requiring large-scale 3D training datasets [139]. By back-propagating scores through differentiable renderers, methods like [140] aggregate 2D gradients across multiple viewpoints to construct coherent 3D score fields, effectively lifting planar priors into volumetric spaces. However, standard SDS often suffers from over-saturation and geometry ambiguities. To address these limitations, recent advancements introduce multi-view consistent diffusion models that jointly generate correlated views rather than optimizing independent projections. For instance, [58] employs a 3D-aware feature attention mechanism to synchronize intermediate states across views within a single reverse process, significantly reducing geometric misalignment. Similarly, [141] integrates correspondence-aware attention layers to facilitate cross-view interactions, enabling holistic scene generation from text or depth conditions.

Beyond static 3D assets, the synthesis of dynamic 4D content (3D + time) introduces the critical challenge of maintaining spatiotemporal consistency, extending the temporal coherence principles established in video generation to volumetric domains. Early attempts extended video diffusion models to 3D by treating dynamic scenes as time-varying neural fields. [142] demonstrated that temporal layers trained on image sequences could be adapted for video generation, a principle subsequently extended to 4D by [143]. This framework combines text-to-video and multi-view diffusion models to optimize dynamic 3D Gaussian Splatting, enforcing both temporal coherence and high-fidelity geometry through a compositional loss strategy. Furthermore, [144] introduces a differentiable “Gaussian flow” concept, linking 3D Gaussian dynamics to pixel velocities to enable direct supervision via optical flow, thereby resolving color drifting issues common in dynamic reconstruction. Efficiency remains a bottleneck; while [139] requires extensive optimization steps, newer approaches like [145] leverage the world simulation capabilities of pretrained video diffusion

models to generate 360-degree orbit frames rapidly, reconstructing high-quality meshes in minutes.

Emerging trends indicate a convergence of explicit geometric control with implicit generative priors to overcome the limitations of purely score-based optimization. Methods such as [146] integrate 3D representations directly into the denoising architecture, enforcing 3D consistency as an inductive bias rather than a post-hoc optimization constraint. Additionally, [147] extends these principles to complex scenes by utilizing locally conditioned diffusion to ensure seamless transitions between semantic parts. Despite these advances, challenges persist in achieving fine-grained controllability over motion dynamics and resolving the ambiguity inherent in monocular inputs. Future research directions point towards unified space-time diffusion architectures, akin to [129], adapted for volumetric data, and the integration of physical laws into the guidance mechanism to ensure physically plausible 4D dynamics. The synthesis of high-fidelity, editable 3D and 4D content thus hinges on bridging the gap between efficient video generation priors and rigorous geometric constraints, setting the stage for the application of these controllable frameworks to domains where physical and anatomical fidelity are paramount, as explored in the subsequent section.

5.5 Domain-Specific Synthesis in Science, Medicine, and Remote Sensing

The application of controllable diffusion and flow-based models to scientific, medical, and remote sensing domains represents a critical frontier where generative fidelity must be reconciled with strict physical and anatomical constraints. Unlike artistic synthesis, where diversity is paramount, these specialized fields demand that generated data adhere to sparse measurements, preserve topological integrity, and satisfy underlying physical laws. In medical imaging, the scarcity of annotated data and privacy regulations have driven the adoption of diffusion models for synthesizing high-quality 2D and 3D volumes. Early works such as [148] and [149] demonstrated that Denoising Diffusion Probabilistic Models (DDPMs) superiorly capture the complex distributions of MRI and CT scans compared to GANs, achieving higher fidelity and diversity. However, unconditional generation often fails to guarantee anatomical correctness. To address this, [150] introduced segmentation-guided mechanisms that enforce multi-class anatomical masks at each sampling step, ensuring generated images strictly follow user-defined structural priors. Similarly, [151] integrates lightweight spatial control into pre-trained models to precisely locate pathologies without fine-tuning, significantly reducing anatomical hallucinations.

Beyond static image synthesis, controllable models are pivotal for longitudinal analysis and modality translation. [152] adapts diffusion processes to generate intermediate temporal volumes in 4D cardiac data, learning spatial deformations along geodesic paths to preserve topology across time. For multi-modal synthesis, [153] operates in latent space with cooperative filtering to balance multiple input modalities while preserving anatomical structures via brain region masks. Furthermore, the generation of synthetic data for downstream perception tasks has gained traction; [154] and [155] exploit cross-attention maps to automatically generate pixel-perfect segmentation masks alongside images, drastically reducing annotation costs. Privacy preservation is another critical aspect, with [156] showing that differentially private fine-tuning can produce synthetic medical data that maintains utility for downstream classifiers while providing rigorous privacy guarantees.

In remote sensing and environmental modeling, the challenge lies in generating geographically plausible scenes from sparse or low-resolution inputs. [157] proposes a framework tailored for remote sensing that integrates global and local control inputs to ensure generated imagery contains accurate geographic and temporal information. Extending this to three dimensions, [158] utilizes diffusion models to generate texture colors for point-level geometries derived from satellite imagery,

enabling the rendering of arbitrary views with inter-frame consistency. The integration of physical constraints is also emerging as a vital trend. [50] reframes sampling as a constrained optimization problem, employing iterative projections to ensure outputs adhere to non-convex constraints and ordinary differential equations derived from physical principles. Additionally, [106] extends guidance mechanisms to discrete state spaces, enabling controllable generation of molecular graphs and DNA sequences, which is essential for drug discovery and genomics.

Despite these advancements, significant challenges remain in balancing controllability with the stochastic nature of diffusion processes. Current approaches often rely on heavy conditional guidance that can compromise sample diversity or require extensive retraining for domain adaptation. Future research must focus on developing unified flow-matching frameworks that can seamlessly integrate heterogeneous scientific constraints—ranging from biomechanical laws in medicine to radiative transfer equations in remote sensing—into the generative trajectory without sacrificing sampling efficiency. The convergence of neuro-symbolic reasoning with diffusion priors offers a promising direction to ensure that generated scientific data is not only visually realistic but also logically and physically consistent.

6 Efficiency Optimization, Evaluation, and Ethical Considerations

6.1 Acceleration Architectures and Distillation Paradigms

The prohibitive computational latency inherent in iterative sampling remains a primary bottleneck for the real-time deployment of controllable diffusion and flow-based models. To address this, recent advancements have coalesced around three synergistic paradigms: trajectory rectification, consistency distillation, and architectural streamlining. Theoretical unification via Flow Matching has been instrumental, demonstrating that learning straight probability paths significantly reduces the curvature of generative trajectories, thereby minimizing the discretization error of numerical solvers. Specifically, Rectified Flow frameworks [10] and Flow Matching for Generative Modeling [9] reformulate the generation process as an optimal transport problem, enabling high-fidelity synthesis with drastically fewer function evaluations compared to traditional stochastic differential equation solvers. This rectification is further enhanced by recursive reflow procedures that progressively straighten paths, allowing for accurate simulation with coarse time steps [159].

Parallel to trajectory optimization, consistency distillation has emerged as a dominant strategy for achieving one-step or few-step generation. Consistency Models [160] impose a self-consistency constraint across different noise levels, allowing the model to map any point on a probability flow trajectory directly to its endpoint. When applied to latent spaces, Latent Consistency Models (LCMs) [161] enable rapid inference on pre-trained backbones like Stable Diffusion, achieving state-of-the-art performance in as few as two to four steps. Building on this, SDXS [162] combines knowledge distillation with model miniaturization to achieve real-time frame rates exceeding 100 FPS, while retaining compatibility with image conditions. Furthermore, piecewise rectification strategies such as PeRFlow [45] offer a universal acceleration mechanism by straightening trajectories within specific time windows, preserving the transferability of pre-trained weights.

Architectural innovations have also targeted the overhead of conditional injection mechanisms. While ControlNet [4] established the standard for spatial control via zero-convolutions, its computational cost scales linearly with the number of conditions. Uni-ControlNet [5] addresses this by unifying multiple control modes into a constant number of adapters, significantly reducing parameter counts. More recently, ControlNeXt [53] replaces heavy auxiliary branches with efficient Cross Normalization, reducing learnable parameters by up to 90% without compromising control fidelity.

In the video domain, efficiency is further pursued through cascade architectures like NUWA-XL [163], which employs a “diffusion over diffusion” strategy to generate long sequences in parallel, and Motion-I2V [130], which factorizes motion prediction to decouple temporal complexity.

Despite these strides, a critical trade-off persists between sampling speed and the precision of fine-grained control. Aggressive distillation often leads to mode collapse or reduced adherence to complex structural constraints, particularly in high-resolution regimes. Future research must therefore focus on adaptive sampling schedules that dynamically balance trajectory linearity with constraint satisfaction, and the development of hardware-aware architectures, such as Diffusion Mamba [76; 77], that leverage linear-complexity state space models to scale controllability without quadratic attention overhead.

6.2 Comprehensive Evaluation Frameworks for Controllability and Alignment

Bridging the gap between the efficiency advancements discussed previously and the security challenges addressed subsequently requires a rigorous re-evaluation of how controllable diffusion and flow-based models are assessed. The evaluation paradigm must shift from measuring unconditional image fidelity to quantifying the precise alignment between heterogeneous control signals and generated outputs. Traditional metrics such as Fréchet Inception Distance (FID) and Inception Score, while effective for measuring sample quality and diversity [1; 164], fail to capture the nuanced adherence to structural constraints or semantic layouts inherent in conditional synthesis. A comprehensive framework must therefore decompose performance into distinct dimensions: structural fidelity, semantic consistency, and prompt sensitivity, each critical for validating the trade-offs between speed and control fidelity noted in prior sections.

Structural adherence is paramount in tasks requiring spatial precision, such as layout-to-image generation or edge-guided synthesis. Recent works have introduced specialized metrics to evaluate the Intersection over Union (IoU) between generated segmentation maps and input conditions, or the structural similarity (SSIM) against ground truth geometry. For instance, [126] demonstrates significant improvements in Controllable Average Score (CAS), highlighting the necessity of object-aware cross-attention mechanisms for precise spatial control. Similarly, [54] proposes optimizing pixel-level cycle consistency via discriminative reward models, reporting substantial gains in mIoU and RMSE for depth and edge conditions. These quantitative measures are complemented by attention-based analyses, where [165] leverages cross-attention map manipulation to ensure attribute binding, validating improvements through head-to-head user comparisons that reveal limitations in automated metrics.

Beyond structural constraints, evaluating semantic alignment and multi-subject coherence remains challenging, particularly as models strive for higher fidelity without collapsing into mode collapse—a risk exacerbated by the aggressive distillation techniques mentioned earlier. Methods like [62] address semantic leakage by bounding information flow, yet standard metrics often overlook the subtle blending of identities in complex scenes. To mitigate this, [101] introduces the CMIGBench benchmark, utilizing character-character and text-image similarity scores to assess consistency across multi-turn interactions. Furthermore, [122] establishes DragBench to evaluate pixel-level precision in interactive editing, emphasizing the gap between global image quality and local edit accuracy. The emergence of retrieval-augmented and open-vocabulary approaches, as seen in [166], further complicates evaluation by requiring metrics that can assess alignment with tokens absent from the original training vocabulary.

Crucially, a robust evaluation framework must account for perceptual variability and human prefer-

ence, serving as a necessary precursor to assessing the potential for malicious manipulation. [167] underscores the importance of aligning model outputs with human aesthetic expectations, noting that noise schedule adjustments significantly impact perceived realism. Similarly, [168] integrates identity consistency and aesthetic reward fine-tuning, demonstrating that automated scores often diverge from human judgment in identity preservation tasks. The [169] proposes the LMM Score within the EditEval benchmark to systematically evaluate text-guided editing, bridging the gap between automated assessment and human-centric criteria.

Future evaluation protocols must evolve towards unified benchmarks that standardize inference conditions across diverse modalities, as advocated by [5]. The integration of Large Language Models for grounded generation, as explored in [170], suggests that future metrics should also quantify the model’s reasoning capabilities regarding complex spatial and numerical prompts. Ultimately, the field requires a holistic scoring system that weights structural adherence, semantic fidelity, and aesthetic quality dynamically based on the specific control modality, moving beyond static, one-size-fits-all indices. Such rigorous standardization is essential not only for benchmarking progress but also for establishing the baseline fidelity required to detect the adversarial vulnerabilities and forensic anomalies explored in the following section.

6.3 Security Vulnerabilities, Forensics, and Defensive Mechanisms

The deployment of controllable diffusion and flow-based models introduces profound security challenges, stemming from their dual-use potential for both creative synthesis and malicious manipulation. A primary vulnerability lies in the susceptibility of these models to adversarial perturbations and backdoor injections. Recent investigations reveal that text-to-image diffusion models can be easily compromised through multimodal data poisoning, where attackers inject triggers at pixel, object, or style levels to manipulate generation outcomes without degrading benign utility [171]. Furthermore, the iterative denoising process is vulnerable to adversarial supervision; while robust classifiers can enhance layout alignment [172], non-robust gradients may be exploited to steer trajectories toward semantically incorrect or harmful manifolds [22]. Beyond external attacks, internal privacy risks are acute; diffusion models have been shown to memorize training data, allowing adversaries to extract copyrighted logos or private faces via generate-and-filter pipelines, a vulnerability significantly more pronounced than in prior GAN architectures [173].

Forensic analysis of synthetic media generated by these models presents distinct challenges due to their unique spectral signatures. Unlike GANs, diffusion models leave specific artifacts in the frequency domain, characterized by anomalous regular patterns in autocorrelation and distinct radial spectral power distributions [174]. While traditional detectors struggle with the high photorealism of diffusion outputs, novel frameworks leveraging Diffusion Reconstruction Error (DIRE) have emerged, exploiting the observation that real images cannot be accurately reconstructed by a pre-trained diffusion model whereas synthetic ones can [175]. Similarly, latent space reconstruction errors provide efficient features for distinguishing fake from real content with superior speed and accuracy [176]. However, these forensic capabilities are increasingly threatened by anti-forensic techniques; methods like StealthDiffusion (inferred from context of adversarial evasion in provided literature such as [177]) and transferable evasion strategies enable the generation of imperceptible perturbations that bypass detection systems and content safety filters.

Defensive mechanisms have evolved to address these threats through both model hardening and data protection. To safeguard intellectual property and prevent unauthorized training, techniques such as “unlearnable examples” inject protective noise into source images, rendering them ineffective for

model learning [177]. For copyright authentication post-generation, Contrasting Gradient Inversion (CGI-DM) offers a novel approach by maximizing the KL divergence between pretrained and fine-tuned models to visualize infringement indicators [178]. Additionally, differentially private fine-tuning has proven effective in generating useful synthetic data while providing rigorous privacy guarantees, significantly reducing membership inference risks [156]. Despite these advances, a critical tension remains between controllability and security; enhancing conditional adherence often increases the attack surface for backdoors [171]. Future research must therefore focus on developing unified frameworks that integrate robust certification mechanisms directly into the flow matching objectives, ensuring that the deterministic paths of rectified flows [71] remain immune to trajectory hijacking while preserving the fidelity required for high-stakes applications in medicine and forensics.

6.4 Ethical Stewardship, Bias Mitigation, and Responsible Deployment

Building upon the technical security challenges and defensive mechanisms outlined previously, the deployment of highly controllable diffusion and flow-based models necessitates a rigorous examination of their broader societal implications, particularly regarding the amplification of biases, the potential for malicious manipulation, and the governance of synthetic media. As these models transition from unconditional generation to precise user-directed control, the risk of propagating stereotypical associations through conditional inputs becomes paramount. Research indicates that models trained on unfiltered web-scale datasets often exhibit degenerated behaviors, reinforcing biases related to gender, race, and profession when conditioned on specific prompts. To mitigate this, approaches such as Safe Latent Diffusion (SLD) have been introduced to suppress inappropriate content during the diffusion process without requiring retraining, effectively filtering harmful concepts while preserving image quality and text alignment [179]. Furthermore, methods like Fair Mapping and manifold guidance aim to actively correct demographic imbalances, ensuring that control signals do not exacerbate existing societal inequities.

Beyond bias, the ethical boundaries of image editing and the potential for deepfake creation demand robust defensive mechanisms that complement the forensic and privacy protections discussed earlier. The ability to perform precise, text-guided edits on real images, as demonstrated by frameworks like Imagic and DiffEdit, raises concerns regarding identity fraud and misinformation. Consequently, the development of reliable detection and provenance systems is critical. Forensic analysis has revealed that diffusion models leave unique spectral anomalies and frequency domain discrepancies distinct from real images, which can be exploited for detection [174; 180]. Techniques such as Diffusion Reconstruction Error (DIRE) and Latent Reconstruction Error (LaRE²) leverage the inability of diffusion models to perfectly reconstruct real images compared to synthetic ones, providing robust metrics for distinguishing authentic media from generated artifacts [175; 176]. Additionally, privacy vulnerabilities remain a significant concern, as studies have shown that diffusion models can memorize and emit individual training samples, including sensitive personal data and trademarked logos. This underscores the necessity for differentially private training protocols to generate useful synthetic data while guaranteeing privacy preservation [173; 156].

To ensure responsible deployment, the integration of watermarking and content authentication into the generative pipeline is essential as a proactive measure alongside reactive detection. Recent advancements propose end-to-end optimization techniques to embed imperceptible, verifiable watermarks into generated images, facilitating attribution and tracking across distribution channels [181]. Moreover, the interpretability of latent spaces offers a pathway for safety interventions; by identifying directions associated with harmful concepts, researchers can implement precise constraints to prevent unsafe generation without degrading model utility [182]. Looking forward, ethical stew-

ardship must evolve from post-hoc mitigation to proactive design, incorporating human-in-the-loop feedback systems and regulatory standards that govern the use of controllable synthesis in sensitive domains such as journalism and healthcare. The community must prioritize the development of unified benchmarks that evaluate not only fidelity and controllability but also fairness, robustness against adversarial attacks, and adherence to ethical guidelines, ensuring that the transformative power of these models benefits society while minimizing potential harms.

7 Conclusion

This subsection synthesizes the paradigmatic shift from unconditional synthesis to precise, user-directed control within diffusion and flow-based generative frameworks. The survey has delineated a trajectory where controllability has evolved from coarse classifier guidance [21] to fine-grained architectural interventions such as adapter-based spatial conditioning [53; 5] and attention-centric modulation [7; 62]. While current methodologies have achieved remarkable fidelity in aligning visual outputs with heterogeneous signals—including text, layout maps, and reference images—significant trade-offs persist between sampling efficiency, control precision, and model generalizability. For instance, while rectified flows and consistency models offer accelerated sampling trajectories [77; 183], they often struggle to maintain the nuanced adherence to complex logical constraints observed in slower, iterative solvers. Furthermore, the compositional limitations of monolithic models, particularly in binding multiple attributes to distinct objects, remain a critical bottleneck, despite advances in energy-based composition [103; 85] and neuro-symbolic planning [184; 170].

A persistent gap lies in the theoretical disconnect between the stochastic dynamics of diffusion processes and the deterministic requirements of strict constraint satisfaction. Although frameworks like Projected Generative Diffusion Models [50] and D-Flow [34] attempt to bridge this by recasting generation as constrained optimization or differentiable flow inversion, a unified theory that guarantees both physical plausibility and semantic alignment across arbitrary modalities is yet to be established. Current evaluation metrics predominantly focus on perceptual quality and prompt alignment, often neglecting the rigorous quantification of structural adherence and causal consistency. This lack of standardized benchmarks hampers the objective comparison of emerging techniques, such as reward-based fine-tuning [48; 168] versus training-free inference strategies [185].

Looking forward, the field must pivot towards generative systems endowed with explicit causal reasoning and neuro-symbolic integration. Future architectures should transcend pattern matching to incorporate world models capable of understanding physical laws and temporal dynamics, thereby enabling robust spatiotemporal control in video and 3D synthesis [12; 135; 186]. The integration of Large Language Models as high-level planners [187; 70] offers a promising avenue for decomposing complex instructions into executable visual operations, yet achieving seamless, real-time human-in-the-loop interaction requires further advancements in efficient, universal interfaces [6; 75]. Ultimately, the democratization of high-fidelity creative tools hinges on resolving these theoretical and architectural challenges while simultaneously enforcing rigorous ethical stewardship. As models become increasingly capable of manipulating reality, ensuring robustness against adversarial exploitation and mitigating bias amplification [179] remains paramount to fostering responsible innovation in controllable visual synthesis.

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